

DÚ LA 3.2

(a) Provedeme Gram-Schmitovu ortogonalizaci na posloupnost $(1, x, x^2)$ (značení vektorů je stejné jako ve skriptech).

1. $\mathbf{v}_1 = 1$

$$\|\mathbf{v}_1\| = \sqrt{\int_1^5 1 \, dx} = 2$$

$$\mathbf{u}_1 = \frac{1}{2} = 0,5$$

2. $\mathbf{v}_2 = x$

$$\mathbf{w}_2 = \langle \frac{1}{2}, x \rangle \frac{1}{2} = \frac{1}{2} \int_1^5 \frac{x}{2} \, dx = 3$$

$$\|\mathbf{v}_2 - \mathbf{w}_2\| = \sqrt{\int_1^5 (x-3)^2 \, dx} = \frac{4}{\sqrt{3}}$$

$$\mathbf{u}_2 = \frac{\sqrt{3}}{4}(x-3) \approx 0,433x - 1,299$$

3. $\mathbf{v}_3 = x^2$

$$\mathbf{w}_3 = \langle \frac{1}{2}, x^2 \rangle \frac{1}{2} + \langle \frac{\sqrt{3}}{4}(x-3), x^2 \rangle \frac{\sqrt{3}}{4}(x-3) = \frac{1}{2} \int_1^5 \frac{x^2}{2} \, dx + \frac{\sqrt{3}}{4}(x-3) \int_1^5 \frac{\sqrt{3}}{4}(x^3 - 3x^2) \, dx \approx 6,000x - 7,667$$

$$\|\mathbf{v}_3 - \mathbf{w}_3\| \approx \sqrt{\int_1^5 (x^2 - 6,000x + 7,667)^2 \, dx} \approx 2,385$$

$$\mathbf{u}_3 \approx 0,419x^2 - 2,516x + 3,214$$

(b) Označme $\mathbf{w}$ hledanou OG projekci $\sin x$ do $\text{Span}\{1, x, x^2\}$. Pro výpočet využijeme tvrzení 8.54 a ON bázi podprostoru nalezenou výše.

$$\mathbf{w} = \langle \mathbf{u}_1, \sin x \rangle \mathbf{u}_1 + \langle \mathbf{u}_2, \sin x \rangle \mathbf{u}_2 + \langle \mathbf{u}_3, \sin x \rangle \mathbf{u}_3 \approx \frac{1}{4} \int_1^5 \sin x \, dx + (0,433x - 1,299) \int_1^5 \sin x \, dx + (0,419x^2 - 2,516x + 3,214) \int_1^5 (0,419x^2 - 2,516x + 3,214) \sin x \, dx \approx -0,053x^2 - 0,331x + 1,601$$

(c) $\|\sin x - (-0,053x^2 - 0,331x + 1,601)\| = \sqrt{\int_1^5 (\sin x + 0,053x^2 + 0,331x - 1,601)^2 \, dx} \approx 0,319$